

# Hossein Baktash

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## Education

- 2021 – now **Ph.D. in Electrical and Computer Engineering**, *Carnegie Mellon University*, Pittsburgh, PA, *advisor*: Keenan Crane.
- 2015–2020 **B.Sc. in Computer Engineering**, *Sharif University of Technology*, Tehran, Iran.
  - *Minor in Mathematics*
- 2014–2015 **Team selection camp for International Mathematical Olympiad**, *Young Scholars Club*, Tehran, Iran.
- 2011–2014 **Diploma in Mathematics and Physics Discipline**, *Shahid Soltani High School (NODET)*, Karaj, Iran.

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## Publications

### –Subgrid Marching Tetrahedra

**Hossein Baktash**, Mark Gillespie, Keenan Crane  
ACM Transactions on Graphics (SIGGRAPH 2026).

### –Putting Rigid Bodies to Rest

**Hossein Baktash**, Nicholas Sharp, Qingnan Zhou, Keenan Crane, Alec Jacobson  
ACM Transactions on Graphics (SIGGRAPH 2025).

### –Computational Imaging using Ultrasonically-Sculpted Virtual Lenses

**Hossein Baktash**, Yash Belhe, Matteo Giuseppe Scopelliti, Yi Hua, Aswin Sankaranarayanan, Maysam Chamanzar  
IEEE Intl. Conf. Computational Photography (ICCP 2022).

### –Some Results on Dominating Induced Matching

S. Akbari, A. Bahjati, H. Baktash, A. Behmaram, M. Roghani  
Graphs and Combinatorics, 2022.

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## Awards and Honors

- 2021 **Carnegie Institute of Technology Dean's Fellowship**, *Carnegie Mellon University*.
- 2014 **Iranian National Mathematics Olympiad**, *Gold medal*, Young Scholars Club, Tehran, Iran.

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## Research Experience

- summer 2026 **AI Scientist Intern**, *Roblox*, San Mateo, CA.

- summer 2023 **Research Scientist Intern**, *Adobe*, NYC.  
 & 2024 Supervised by Alec Jacobson and Qingnan Zhou. Developed an algorithm to identify resting probabilities of rigid bodies and used it for inverse design of shapes with irregular dice behavior.
- 2022–now **Graduate Researcher**, *Geometry Collective*, Carnegie Mellon University.  
 Advised by Keenan Crane. Working on various novel discrete representations for shapes used in geometry processing.
- 2021–2022 **Graduate Researcher**, *Chamanzar Lab*, Carnegie Mellon University.  
 Developed an image processing pipeline to relay and deblur images using unconventional ultrasonic lenses in water.
- 2019–2020 **Bachelor Thesis**, *Sharif University of Technology, Tehran*.  
 Worked on recovering the structure of Ising Blockmodels with more than two blocks. Improved the immediate sample complexity of  $\mathcal{O}(n^2 \log(n))$  to the information theoretic lower bound of  $\mathcal{O}(n \log(n))$  using a variant of the max-cut SDP relaxation.
- summer 2019 **Research Internship**, *INRIA - I3S Laboratoire, France*.  
 Studied several pruning and hashing methods to compress neural networks including Hashed-Net, OBD, and magnitude-based pruning with  $\ell_1$  and  $\ell_2$  regularization methods.

## Talks

### Putting Rigid Bodies to Rest

Invited Talk, Dynamic Graphics Project (DGP), University of Toronto, 2025 (remote)

### Computational Imaging using Ultrasonic Lenses

ICCP 2022, Caltech, Pasadena, CA

## Teaching Experiences

### Teaching Assistant.

- CMU: Discrete Differential Geometry, Mathematical Foundations of Electrical Engineering
- SUT: Linear Algebra, Probability and Statistics, Theory of Languages and Automata

### Tutoring.

- Geometry, Combinatorics, Number Theory for Math/Informatics Olympiad participants

## Skills

**programming** C++, Python, MATLAB, Java

**Tools** Blender, Illustrator, Git, Geogebra, Slurm

**Languages** Persian, English, Azeri

## Additional Activities

Sports & Hobbies Mountaineering, Horse Riding, Snooker/Pool, Drawing, Cooking

- Executive experience
- **Member of Sharif University's Mountaineering Group (2016–2020)**  
Assigned as team leader several times and successfully carried out the responsibilities, which often involved leading a team of 20-30 members in nature, outdoors, and remote regions of the country.
  - **Team guide in IOI 2017 (International Olympiad in Informatics)**  
I was honored to be the guide of team Latvia in the competition. Providing real-time translation and helping the team enjoy their one-week stay in Tehran.